NORTHWIND TRADERS BUSINESS INTELLIGENCE DASHBOARD REPORT

HNG Data Analysis Stage 3 Submission

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Visualization Tool Used: Power BI

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1. Introduction

Northwind Traders is a global gourmet food supplier that operates across multiple countries, managing customers, products, employees, and shipping operations. The objective of this project is to design and develop an interactive Business Intelligence dashboard that provides actionable insights into the company's sales performance, product dynamics, regional distribution, and operational efficiency.

Using the provided Northwind relational dataset, this project transforms raw transactional records into executive-level analytical visuals through data cleaning, feature engineering, data modelling, and dashboard design in Power BI.

The completed dashboard is structured into three major analytical sections:

- Sales & Revenue Overview
- Product & Category Intelligence
- Regional, Operational & People Performance

This project is aimed at helping executives make informed, data-driven business decisions.

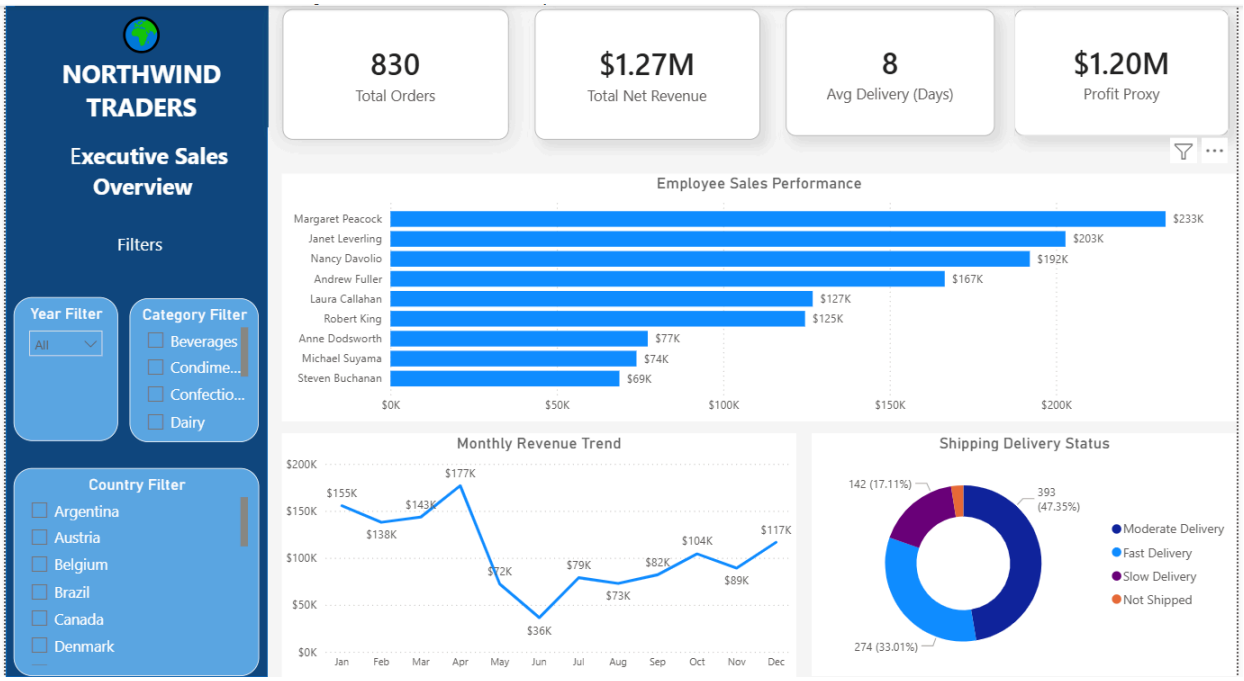

Figure 1: Executive preview of the Northwind Traders Business Intelligence Dashboard

2. Data Sources

The analysis was conducted using seven interconnected CSV files provided as the Northwind Traders dataset. These files represent the key operational dimensions of the business:

- Orders
- Order Details
- Products
- Categories
- Customers
- Employees
- Shippers

Together, these datasets capture the complete sales lifecycle from customer order placement to product shipment and employee sales contribution

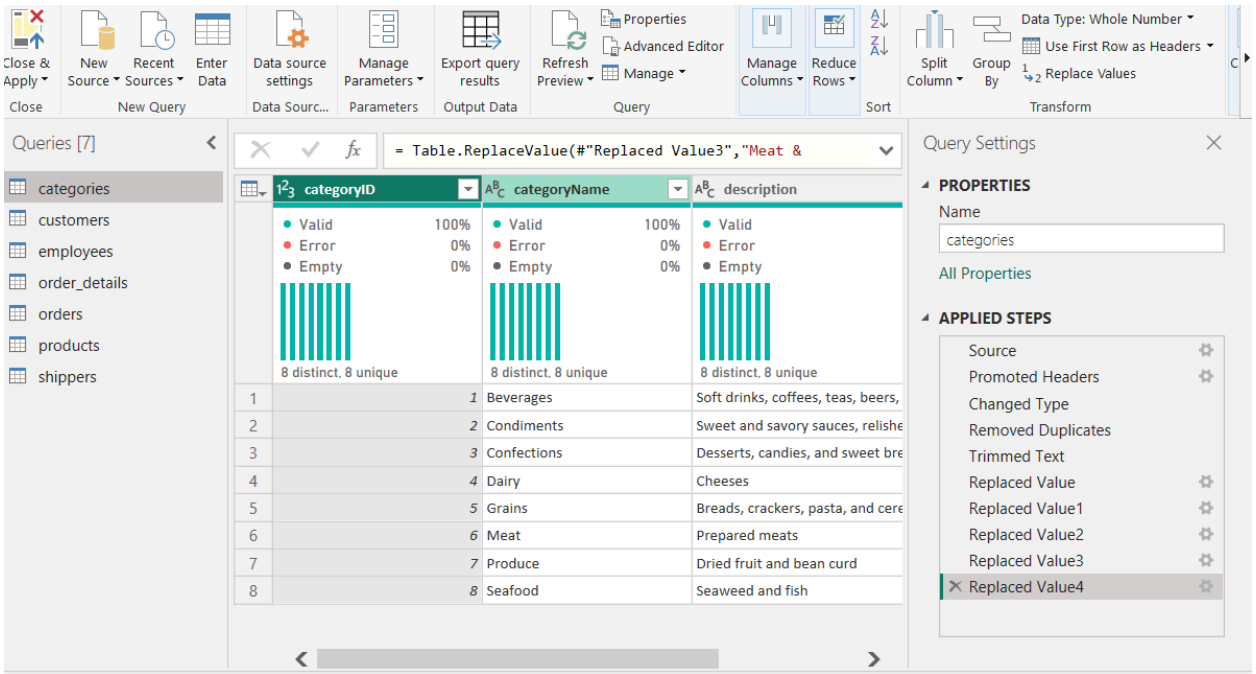

Figure 2: Imported Northwind Traders relational datasets in Power Query Editor.

3. Methodology

This project followed a structured Business Intelligence workflow to ensure that the final dashboard was accurate, interactive, and decision-orientated. The methodology consisted of the following stages:

1. **Data Acquisition:** Importing all CSV files into Power BI Power Query Editor.
2. **Data Validation & Cleaning:** Confirming correct data types, removing duplicates, and handling missing values.
3. **Feature Engineering:** Creating analytical columns for revenue, discounts, delivery performance, and product lifecycle.
4. **Data Modelling:** Building a star schema relationship model between fact and dimension tables.
5. **DAX Measure Development:** Creating key performance indicators and reusable calculations.
6. **Dashboard Design:** Developing three interactive report pages aligned with the required business questions.
7. **Insight Extraction:** Interpreting trends and operational patterns from the completed dashboard.

This methodology ensured a seamless transition from raw data to business intelligence storytelling.

4. Data Cleaning and Preparation

Data cleaning and transformation were performed in Power Query Editor to ensure consistency and reliability across all tables.

4.1 Data Type Standardization

Each field was checked and assigned its proper type:

- Numeric fields (Unit Price, Freight, Discount) → Decimal Number
- ID fields (OrderID, ProductID, EmployeeID) → Whole Number
- Date fields (Order Date, Required Date, Shipped Date) → Date
- Text fields (Country, Category Name, Product Name) → Text

4.2 Duplicate Removal

Duplicate rows were removed from the major transaction tables (`order_details` and `orders`) to ensure accurate aggregation.

4.3 Missing Value Handling

Blank `Shipped Date` values were retained and later classified analytically as **Not Shipped**, rather than deleting potentially important records.

4.4 Text Cleaning

Text columns were trimmed to remove hidden spaces and ensure proper grouping/filtering consistency.

5. Feature Engineering

To enhance analysis, several new calculated columns were created.

5.1 Revenue Columns (order_details)

- Gross Revenue = Unit Price × Quantity
- Discount Amount = Gross Revenue × Discount
- Net Revenue = Gross Revenue – Discount Amount

5.2 Time Intelligence (orders)

- Order Year
- Order Month
- Order Month Number

5.3 Delivery Performance

- Delivery Days = Shipped Date – Order Date
- Delay Status = Fast Delivery / Moderate Delivery / Slow Delivery / Not Shipped

5.4 Product Lifecycle

- Product Status = Active / Discontinued

These engineered columns enabled deeper business insights beyond the raw data.

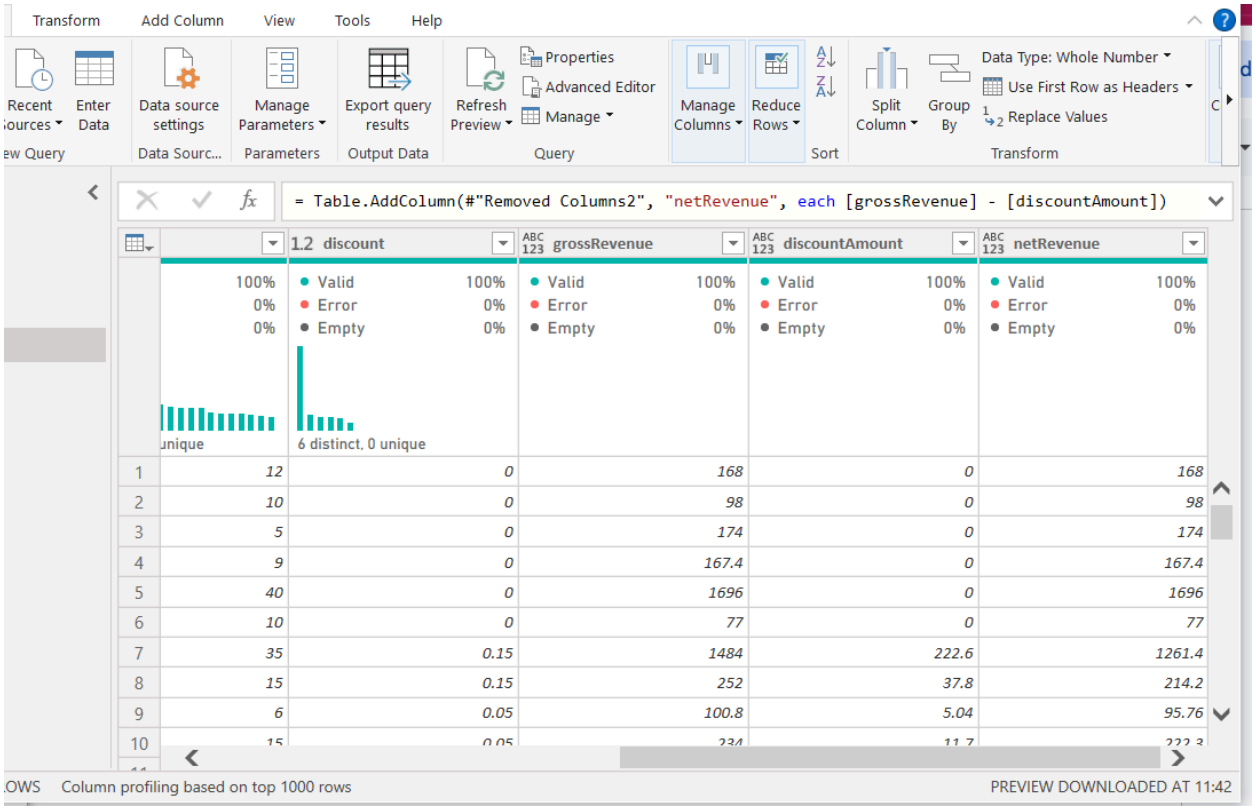

Figure 3: Custom calculated columns created during feature engineering in Power Query.

6. Data Modeling

A star schema data model was developed to support efficient filtering and cross-table calculations.

The `orders` and `order_details` tables served as the transactional fact tables, while `customers`, `employees`, `products`, `categories`, and `shippers` served as supporting dimension tables.

Relationships were configured as:

- Many-to-One cardinality
- Single-direction filtering

This ensured smooth interaction between visuals and slicers across the report.

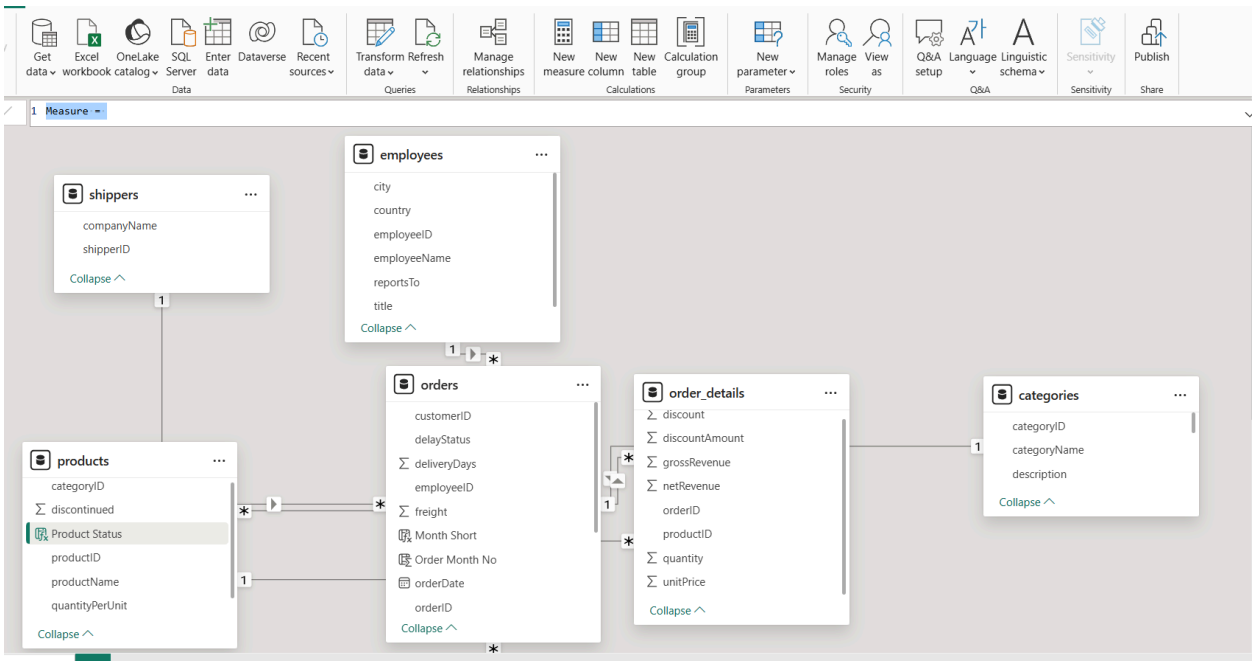

Figure 4: Star schema data model showing fact and dimension table relationships.

7. DAX Measures

The following DAX measures were created to drive KPI cards and visual calculations:

- Total Net Revenue
- Total Orders
- Total Discount
- Total Freight
- Average Delivery Days
- Profit Proxy
- Active Products
- Discontinued Products
- Average Discount Percentage

These measures served as the core business performance indicators across all three dashboard pages.

8. Dashboard Design

The final dashboard was designed as a three-page executive reporting solution.

8.1 Sales & Revenue Overview

This page provides a high-level summary of organisational sales performance.

It answers:

- What are the revenue trends over time?
- What is the total order volume?
- Which employees generate the most revenue?
- How efficient is shipping delivery?

Included Visuals:

- KPI Cards
- Monthly Revenue Trend

- Employee Sales Performance
- Shipping Delivery Status

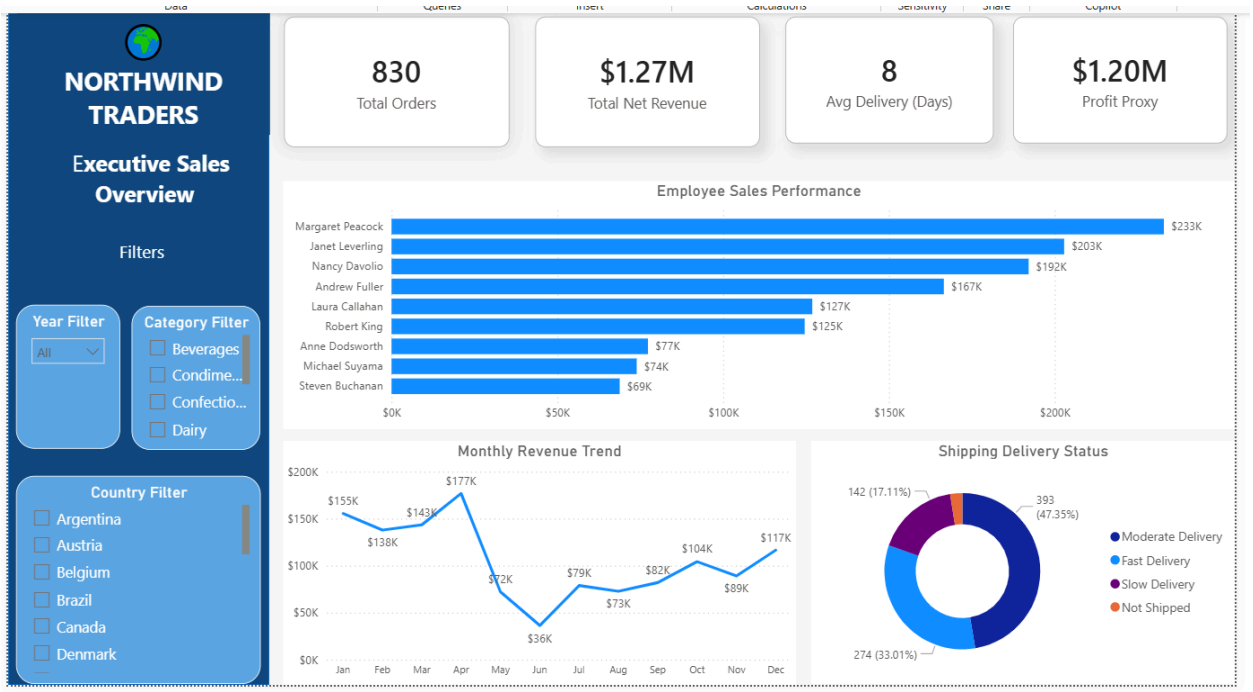

Figure 5: Sales & Revenue Overview dashboard page.

8.2 Product & Category Intelligence

This page analyses product-level business behaviour.

It answers:

- Which product categories generate the highest revenue?
- How do discounts affect profitability?
- What is the proportion of active vs. discontinued products?

Included Visuals:

- Top Revenue Drivers by Category
- Discount vs Profit by Category
- Product Lifecycle Donut Chart

- KPI Cards

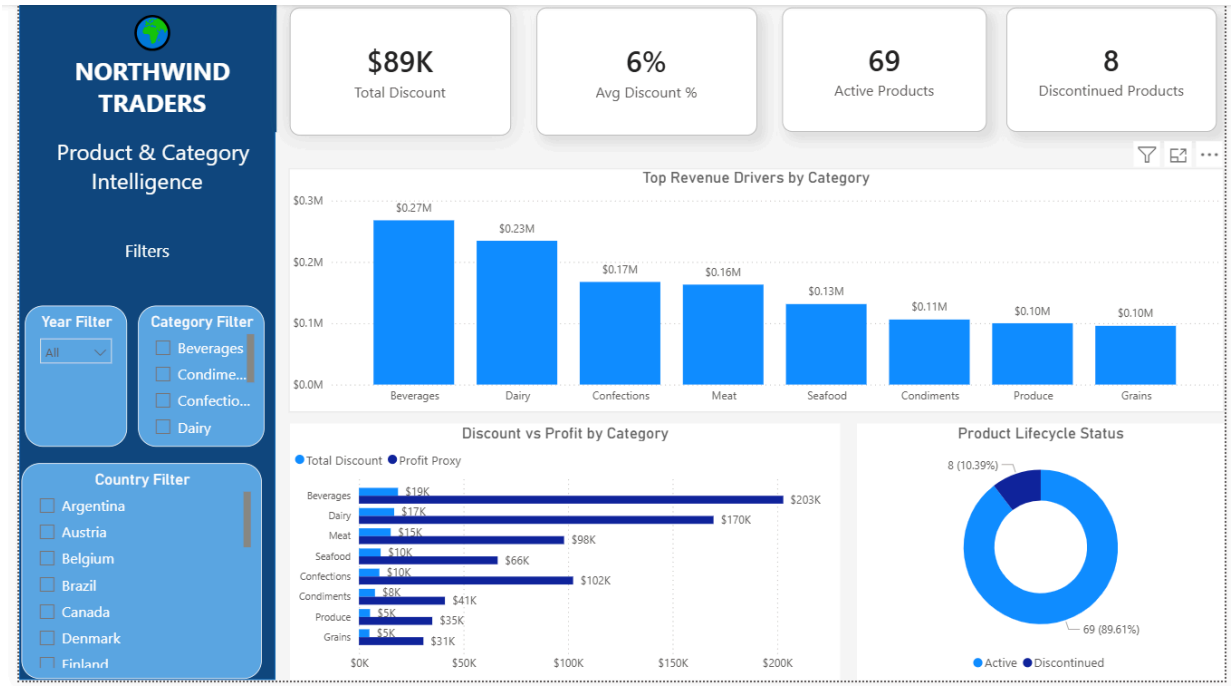

Figure 6: Product & Category Intelligence dashboard page.

8.3 Regional, Operational & People Performance

This page focuses on geographic performance, logistics, and people productivity.

It answers:

- Which countries contribute the highest market share?
- Which shippers are most expensive or slowest?
- How are employees contributing to total revenue?

Included Visuals:

- Top Revenue Markets by Country
- Shipper Cost vs Delivery Speed
- Employee Contribution to Revenue
- KPI Cards

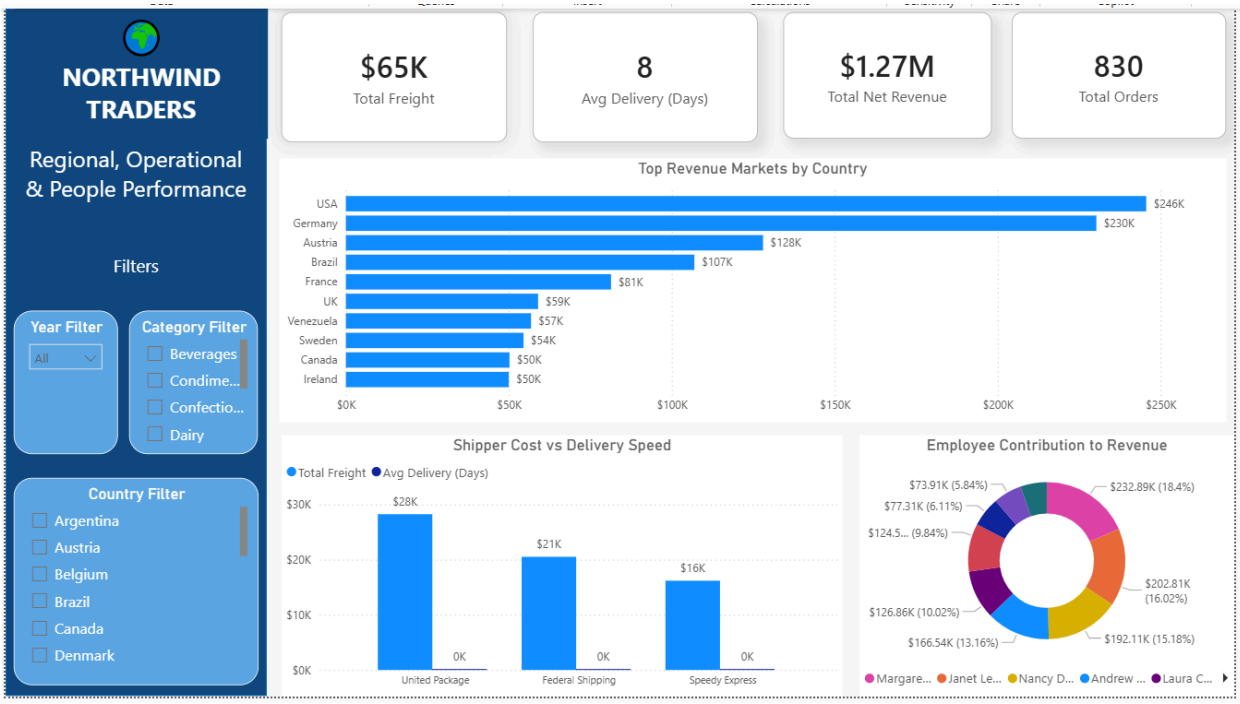

Figure 7: The Regional, Operational & People Performance dashboard page.

9. Filters and Interactivity

To ensure dynamic user exploration, three global slicers were included across all dashboard pages:

- Year Filter
- Category Filter
- Country Filter

These slicers allow executives to instantly drill into specific years, product categories, or markets.

Additionally, chart interactions were configured so that selecting one business dimension dynamically updates the remaining visuals.

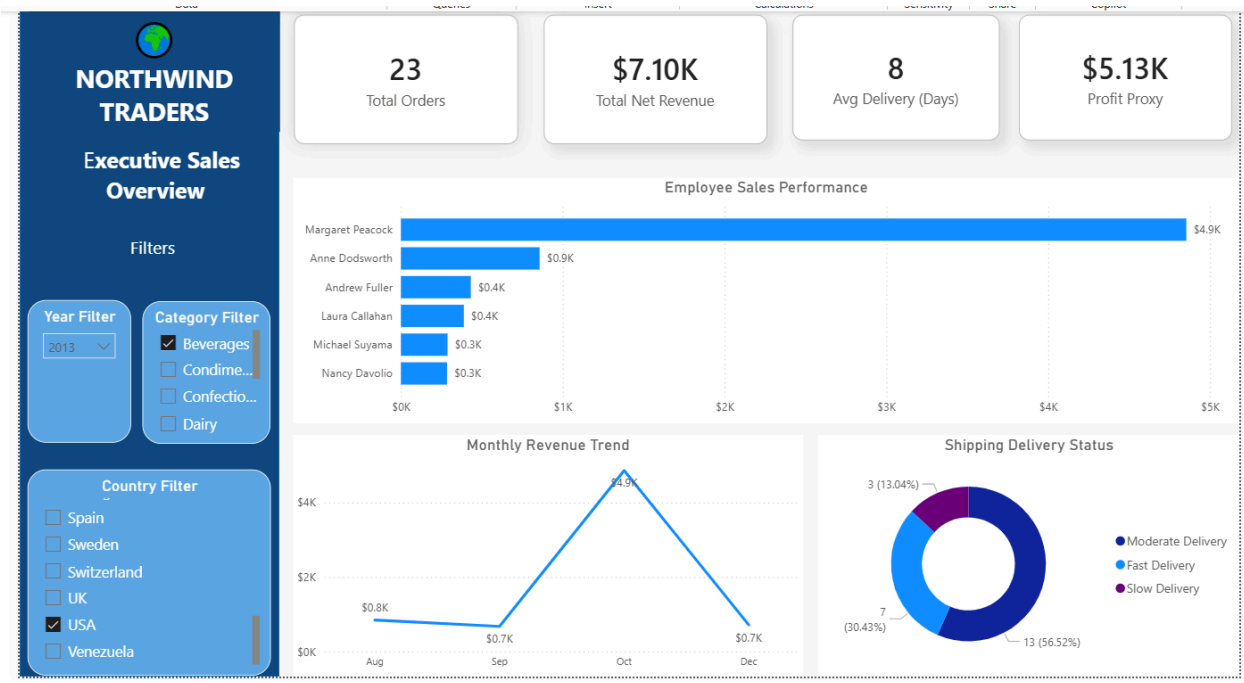

Figure 8: Interactive slicers enabling dynamic filtering across business dimensions.

10. Key Business Insights

From the completed analysis, the following insights were derived:

- Revenue shows noticeable monthly fluctuations, indicating seasonal demand changes.
- A small number of product categories drive a large percentage of total revenue.
- Discounting supports sales volume but weakens retained profitability.
- Certain countries dominate total revenue contribution.
- Freight cost and delivery speed vary significantly by shipper.
- Employee sales contribution is uneven, revealing top-performing staff members.

These insights can support strategic decisions around pricing, logistics optimization, and salesforce management.

11. Conclusion

This project demonstrates the practical application of Power BI in transforming raw business data into an executive-ready Business Intelligence solution.

Through structured cleaning, transformation, modeling, and visualization, Northwind Traders' operational complexity was simplified into a clear analytical dashboard that supports:

- revenue monitoring,
- product strategy,
- regional expansion planning,
- and operational performance tracking.

The resulting dashboard enables data-driven decision-making across multiple business dimensions.

12. Dashboard Shareable Link

Power BI Dashboard Link:

https://app.powerbi.com/view?r=eyJrIjoiOWRjMGQ3MzItNzc2MC00YzMyLWJkODEtMmUxODRmMTk0ZTQ4IiwidCI6IjU2YjNjMjUzLTdlM2MtNDFjNC04NjEwLTc4YjIyYWNjN2E2MCJ9&pageName=4fd98775a21613d9ea76

13. Deliverables Summary

This Google Drive submission folder contains:

- Interactive Power BI Dashboard (.pbix file / shareable report link)
- Project Report (PDF)
- Supporting screenshots and documentation